

BOUNDARY BELYI COVERS AND NORMAL-JET OBSTRUCTIONS IN THE PLANE JACOBIAN PROBLEM

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ABSTRACT. Newton-face differential equations are a classical tool in the plane Jacobian problem. We isolate two structures carried by a face and its formal normal neighborhood.

First, a terminal face equation

$$npq - mzp'q' + nzp'q = 1$$

defines a Belyi map. When the polynomial lattice has gap g , the relevant map is not the ambient degree- mnb cover but its degree- mnb/g quotient. We give its passport, prove automatic connectedness, and compute the first five quotient Hurwitz problems arising from the complete-chain queue: they have 1, 1, 1, 2, 2 classes. Exact formulas through maximum degree 128 are infinitesimally rigid modulo source scaling.

Second, we derive the universal linear operator governing every normal layer of the determinant equation. In relative variables it is a rank-one logarithmic connection plus multiplication by the boundary-volume form. This gives an actual gauge deformation complex. It also gives the adjoint for the complementary layer, an exact residue realization of every finite matrix left kernel, and a filtered one-layer descent criterion. A normalized adjacent contact of type (a, e) , with $e > a^2$, forces the explicit degree- e Belyi map

$$W_{a,e}(s) = \frac{(s-1)^{e-a^2} H_{a,e}(s)^a}{s^e}, \quad H_{a,e} = \frac{1}{e} {}_2F_1\left(-a, 1; 1 - \frac{e}{a}; s\right).$$

Its passport is $(a^a, e - a^2), (e), (a+1, 1^{e-a-1})$. The explicit finite computations use exact arithmetic and are supplied with reproducible certificates. These results organize a terminal boundary-gluing program; they do not prove the plane Jacobian conjecture.

Working computer-assisted manuscript. The recovered exact certificates have been rerun. Independent implementation and specialist review remain submission requirements.

1. INTRODUCTION

The counterexample to the Jacobian conjecture in dimension three leaves the two-variable conjecture open. The separation is geometric: the marked-root construction that fills an affine threefold does not directly fill an affine plane. In the plane, the unresolved information accumulates at infinity in Newton faces, divisorial valuations, and compatibility between successive boundary layers.

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We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the three-dimensional example, and Alpöge announced it publicly [11, 1].

The background here is substantial. One-variable face differential equations of the form used below occur in the work of Formanek and Stothers on the plane Jacobian problem [5]. Borisov developed Belyi maps and dessins in compactification frameworks for hypothetical plane counterexamples [3]. Guccione, Guccione, Horruitiner, and Valqui constructed complete-chain algorithms and reduced the range below maximum degree 125 to the pair (72, 108) and its transpose [6, 7]. Normal-boundary determinant expansions also have a recent valutive precursor in Bado's work [2].

While this manuscript was being prepared, a MathOverflow answer announced a computer-assisted proof that a planar counterexample must have maximum coordinate degree at least 125 [9]. A separate exact-arithmetic campaign by Santibáñez-Leal states strong certificate evidence at the same frontier while distinguishing that evidence from a complete all-coefficient proof [10]. We do not claim the degree-125 bound. The purpose of this paper is to extract reusable boundary theorems from an independent terminal calculation and to state exactly what remains missing from a global argument.

1.1. Main results. The first result corrects a tempting overcount. A terminal face may be written in a fractional uniformizing variable z , but the exponent lattice of the original polynomial pair can force both face polynomials to belong to $k[z^g]$. The actual Hurwitz problem is then a quotient by $u = z^g$.

Theorem 1.1 (Lattice-gap quotient, informal form). *Let $m, n > 1$ be co-prime, let $b \geq 1$, and suppose a terminal face satisfies*

$$npq - mzp'q' + nzp'q = 1, \quad \deg p = mb - 1, \quad \deg q = nb.$$

If $p(z) = \bar{p}(z^g)$ and $q(z) = \bar{q}(z^g)$, as forced by a complete-chain lattice gap g , then $g \mid n$ and

$$\bar{\tau}(u) = u^{n/g} \frac{\bar{p}(u)^n}{\bar{q}(u)^m}$$

is a Belyi map of degree mnb/g , with passport

$$\left(n^{(mb-1)/g}, \frac{n}{g} \right), \quad \left(m^{nb/g} \right), \quad \left(\frac{(m+n)b-1}{g}, 1^{\frac{mnb-(m+n)b+1}{g}} \right).$$

Every permutation triple with this passport is transitive.

Applying the theorem to the first five terminal complete-chain rows gives quotient degrees 6, 10, 9, 9, 16 and respectively 1, 1, 1, 2, 2 dessin classes. The first row is especially useful: an ambient degree-30 passport has eleven classes, but the gap-five lattice selects the unique cyclic pullback of

$$\frac{u(u-1)^5}{(u^2 - \frac{5}{3}u + \frac{5}{9})^3}.$$

The second result describes the normal direction. At layer r , new coefficients enter the determinant equation through a universal first-order operator

$$\mathcal{D}_r^{\alpha,\beta}(a,b) = (\alpha - r)a dB_0 - \beta B_0 da + \alpha A_0 db + (r - \beta)b dA_0.$$

Its formal adjoint is again explicit and converts coefficient obstructions into residue functionals. This is not merely a first-order approximation: the same operator occurs at every normal order, with lower orders appearing only in the forcing term.

Writing $S = \alpha + \beta$, relative variations turn the operator into

$$\mathcal{D}_r(a,b) = A_0 B_0 \left(d + \frac{r}{S} d \log(A_0 B_0) \right) \eta + \frac{S-r}{S} \xi (\alpha A_0 dB_0 - \beta B_0 dA_0).$$

Off resonance, provided the boundary-volume form is nonzero, it is surjective over the function field, so all obstructions come from finite Newton and pole windows. At $r = S$ it becomes an exact differential and its cokernel is controlled by ordinary residues. The full filtered-adjoint statement is proved in appendix A.

The third result gives the adjacent reduced cover forced by a normalized contact.

Theorem 1.2 (Secondary transport, informal form). *For integers $a \geq 1$ and $e > a^2$, set $\delta = e - a^2$. The normalized boundary-volume transport equation has the unique polynomial-numerator solution*

$$h(s) = \frac{H_{a,e}(s)}{(s-1)^a}, \quad H_{a,e}(s) = \sum_{k=0}^a \frac{a^k a!}{(a-k)! \prod_{j=0}^k (e-aj)} s^k.$$

The exceptional ratio

$$W_{a,e}(s) = \frac{(s-1)^\delta H_{a,e}(s)^a}{s^e}$$

is a degree- e Belyi map with passport

$$(a^a, \delta), \quad (e), \quad (a+1, 1^{e-a-1}).$$

For $(a, e) = (4, 17)$, this recovers the degree-17 secondary cover in the terminal (8, 28) calculation. The lesson is negative but useful: reduced boundary covers are often forced and mutually compatible. Any contradiction must live in the global normal-jet gluing, not merely in the first cover.

1.2. Scope. The statements proved here are algebraic theorems about face equations, lattice quotients, normal-layer operators, and the listed finite Hurwitz problems. We do not prove:

- that every terminal complete chain has a complete finite-dimensional boundary deformation model;
- that corrections found separately at each normal layer are compatible;

- that compatible formal corrections integrate to a polynomial transformation using approximate roots; or
- the plane Jacobian conjecture.

Those distinctions will remain visible throughout the paper.

2. FACE EQUATIONS AND THEIR BELYI MAPS

We work over a characteristic-zero field k , embedded in $\mathbb{C}$ when using topological language about covers. A passport of a degree- D Belyi map is the ordered triple of partitions of D recording ramification over the three branch values. The order of the branch values is immaterial for our applications.

2.1. The ambient terminal face. Let $m, n > 1$ be coprime and $b \geq 1$. Consider polynomials

$$p, q \in k[z], \quad \deg p = mb - 1, \quad \deg q = nb,$$

satisfying

$$npq - mzp'q' + nzp'q = 1. \quad (2.1)$$

Define

$$\tau(z) = z^n \frac{p(z)^n}{q(z)^m}. \quad (2.2)$$

Proposition 2.1 (Face-to-passport formula). *The rational function τ is a degree- mnb Belyi map and*

$$\tau'(z) = z^{n-1} \frac{p(z)^{n-1}}{q(z)^{m+1}}. \quad (2.3)$$

Its passport is

$$\left(n^{mb}\right), \quad \left(m^{nb}\right), \quad \left((m+n)b-1, 1^{mnb-(m+n)b+1}\right). \quad (2.4)$$

Proof. Logarithmic differentiation of (2.2) gives

$$\tau' = z^{n-1} \frac{p^{n-1}}{q^{m+1}} (npq + nzp'q - mzp'q').$$

Equation (2.1) yields (2.3).

At $z = 0$, the face equation gives $np(0)q(0) = 1$. At a zero z_0 of p , it gives $nz_0p'(z_0)q(z_0) = 1$; hence the root is simple, nonzero, and not a root of q . The analogous assertion follows for roots of q . Thus the zero fiber of τ has one point of multiplicity n at zero and $mb - 1$ further points of multiplicity n . The pole fiber consists of nb points of multiplicity m .

The numerator and denominator of τ are coprime and both have degree mnb , so $\tau(\infty) \in k^\times$. From (2.3),

$$\tau'(z) = O\left(z^{-(m+n)b}\right) \quad \text{as } z \longrightarrow \infty.$$

Therefore

$$\tau(z) - \tau(\infty) = O\left(z^{-((m+n)b-1)}\right),$$

with nonzero leading coefficient. The ramification contributions already identified total

$$mb(n-1) + nb(m-1) + ((m+n)b-2) = 2mnb - 2.$$

Riemann–Hurwitz leaves no additional critical point. The rest of the $\tau(\infty)$ -fiber is unramified, proving (2.4). $\square$

Remark 2.2. Equation (2.1) is a member of the classical family of one-variable equations studied in the Formanek–Stothers approach [5]. The content of proposition 2.1 is the complete passport forced by the stated endpoint degrees and normalization, not the first appearance of such an equation in the Jacobian problem.

2.2. The lattice quotient. In a complete-chain chart, z can be a fractional uniformizing coordinate. The polynomial exponent lattice forces a gap

$$g = \frac{\rho}{\gcd(\rho, \ell)}$$

and restricts the face polynomials to $k[z^g]$. The following theorem is stated purely algebraically; the complete-chain theory supplies its hypothesis in the applications.

Theorem 2.3 (Lattice-gap terminal cover). *Assume the hypotheses of proposition 2.1 and suppose*

$$p(z) = \bar{p}(u), \quad q(z) = \bar{q}(u), \quad u = z^g,$$

for an integer $g \geq 1$. Then

$$g \mid mb - 1, \quad g \mid nb, \quad \gcd(g, b) = 1, \quad g \mid n.$$

Put

$$\begin{aligned} N &= \frac{n}{g}, & A &= \frac{mb-1}{g}, & B &= \frac{nb}{g}, \\ D &= \frac{mnb}{g}, & H &= \frac{(m+n)b-1}{g}. \end{aligned}$$

The quotient face equation is

$$N\bar{p}\bar{q} - mu\bar{p}\bar{q}' + nu\bar{p}'\bar{q} = \frac{1}{g}, \tag{2.5}$$

and

$$\bar{\tau}(u) = u^N \frac{\bar{p}(u)^n}{\bar{q}(u)^m} \tag{2.6}$$

is a degree- D Belyi map with

$$\bar{\tau}'(u) = \frac{1}{g} u^{N-1} \frac{\bar{p}(u)^{n-1}}{\bar{q}(u)^{m+1}} \tag{2.7}$$

and passport

$$\left(n^A, N\right), \quad \left(m^B\right), \quad \left(H, 1^{D-H}\right). \tag{2.8}$$

Moreover,

$$\tau(z) = \bar{\tau}(z^g),$$

and every permutation triple with passport (2.8) is transitive.

Proof. The degree assumptions imply $g \mid mb - 1$ and $g \mid nb$. The first divisibility gives $\gcd(g, b) = 1$, so the second gives $g \mid n$. Substitution into (2.1) and division by g yield (2.5); logarithmic differentiation gives (2.7). The proof of proposition 2.1, with the zero multiplicity N at $u = 0$, gives the degree and passport. The factorization of τ is immediate.

It remains to prove transitivity. Let $(\sigma_0, \sigma_1, \sigma_\infty)$ have the stated cycle types and product one. An orbit disjoint from the unique H -cycle of σ_∞ is pointwise fixed by σ_∞ . On that orbit, $\sigma_1 = \sigma_0^{-1}$, so its σ_0 - and σ_1 -cycles have equal length. The possible lengths for σ_0 are n and N ; the length for σ_1 is m . The equality $n = m$ contradicts coprimality, while $N = m$ would imply $m \mid n$. Hence no second orbit exists. $\square$

3. THE FIRST QUOTIENT HURWITZ PROBLEMS

The complete-chain algorithm of [6, 7] supplies a finite queue of terminal data at each degree bound. We apply theorem 2.3 to the first five rows at and beyond maximum degree 125.

case	terminal $A; k$	(m, n)	g	quotient	quotient passport	classes
F_2 , 125	$(7/5, 2); 1$	$(3, 5)$	5	6	$(5, 1), (3^2), (3, 1^3)$	1
one-step, 126	$(19/7, 5); 2$	$(2, 3)$	3	10	$(3^3, 1), (2^5), (8, 1^2)$	1
two-step, 126	$(11/6, 3); 1$	$(3, 2)$	2	9	$(2^4, 1), (3^3), (7, 1^2)$	1
F_{24} , 128	$(19/8, 3); 1$	$(3, 4)$	4	9	$(4^2, 1), (3^3), (5, 1^4)$	2
one-step, 132	$(19/4, 8); 1$	$(2, 3)$	3	16	$(3^5, 1), (2^8), (13, 1^3)$	2

TABLE 1. The first lattice-compatible terminal Hurwitz problems.

Proposition 3.1 (Exact Hurwitz count). *The final column of table 1 is exact. In every row the deck group of every connected cover is trivial, so the weighted Hurwitz count equals the ordinary number of isomorphism classes.*

Proof. For conjugacy classes $C_0, C_1, C_\infty \subseteq S_D$, the Frobenius character formula gives the weighted number

$$\frac{|C_0||C_1||C_\infty|}{(D!)^2} \sum_{\chi \in \text{Irr}(S_D)} \frac{\chi(C_0)\chi(C_1)\chi(C_\infty)}{\chi(1)}.$$

The exact verifier evaluates the irreducible characters by the Murnaghan–Nakayama rule. The five resulting rational numbers are 1, 1, 1, 2, 2.

By theorem 2.3, every triple is transitive. In all five rows $N = 1$, so the first branch permutation has a unique fixed sheet. A deck transformation centralizing the transitive monodromy group must preserve that sheet and is therefore the identity. Thus there is no automorphism weight to remove. $\square$

3.1. The degree-125 quotient. For the first row, the ambient passport

$$(5^6), \quad (3^{10}), \quad (15, 1^{15})$$

has eleven connected dessin classes: eight with trivial deck group, two with deck group C_3 , and one with deck group C_5 . The gap is five, so only the last class descends to the polynomial lattice.

Proposition 3.2. *Up to source and target normalization, the unique lattice-compatible quotient is obtained from*

$$\bar{p} = 1 - u, \quad \bar{q} = \frac{1}{5} - \frac{3}{5}u + \frac{9}{25}u^2.$$

It satisfies

$$\bar{p}\bar{q} - 3u\bar{p}\bar{q}' + 5u\bar{p}'\bar{q} = \frac{1}{5},$$

and its Belyi map is

$$\bar{\tau}(u) \doteq \frac{u(u-1)^5}{\left(u^2 - \frac{5}{3}u + \frac{5}{9}\right)^3}. \quad (3.1)$$

The ambient face is the cyclic pullback $\bar{\tau}(z^5)$.

Proof. Substitution proves the displayed differential equation. The derivative identity and passport follow from theorem 2.3. The exact character and quotient-orbit enumeration proves uniqueness. $\square$

3.2. Explicit post-125 covers. For the one-step degree-126 row, put

$$P = u^3 + u^2 + \frac{5}{12}u + \frac{1}{18},$$

$$Q = u^5 + \frac{3}{2}u^4 + u^3 + \frac{1}{3}u^2 + \frac{5}{96}u + \frac{1}{576}.$$

The exact identity

$$uP^3 - Q^2 = -\frac{36u^2 + 28u + 9}{2985984} \quad (3.2)$$

gives $\bar{p} = 18P$, $\bar{q} = 192Q$.

For the two-step degree-126 row, one may take

$$\bar{p} = 1 + \frac{20}{3}u + 24u^2 + \frac{288}{7}u^3 + \frac{288}{7}u^4,$$

$$\bar{q} = \frac{1}{2} + 5u + 12u^2 + 18u^3.$$

Equivalently,

$$P = u^4 + u^3 + \frac{7}{12}u^2 + \frac{35}{216}u + \frac{7}{288}, \quad Q = u^3 + \frac{2}{3}u^2 + \frac{5}{18}u + \frac{1}{36}$$

satisfy

$$uP^2 - Q^3 = -\frac{72u^2 + 39u + 16}{746496}. \quad (3.3)$$

For F_{24} there are two conjugate solutions over $\mathbb{Q}(\sqrt{6})$. With $\varepsilon = \pm 1$,

$$\begin{aligned}\bar{p}_\varepsilon &= 1 + u + \left(\frac{1}{3} + \varepsilon \frac{\sqrt{6}}{18}\right) u^2, \\ \bar{q}_\varepsilon &= \frac{1}{4} + \frac{5}{8}u + \left(\frac{2}{5} + \varepsilon \frac{\sqrt{6}}{40}\right) u^2 + \left(\frac{17}{160} + \varepsilon \frac{11\sqrt{6}}{480}\right) u^3.\end{aligned}$$

Direct substitution verifies (2.5) in each case.

3.3. Infinitesimal rigidity. Fix the constant terms of a quotient solution and consider

$$\mathcal{F}(p, q) = Npq - mupq' + nup'q.$$

On coefficient variations with zero constant terms, its differential is

$$\mathcal{L}(\alpha, \beta) = N(\alpha q + p\beta) - mu(\alpha q' + p\beta') + nu(\alpha'q + p'\beta). \quad (3.4)$$

Source scaling gives the kernel vector

$$(\alpha, \beta) = (up', uq').$$

Proposition 3.3 (Rigidity of the explicit reduced covers). *For the degree-6, degree-10, degree-9, and two conjugate degree-9 maps above, $\mathcal{L}$ is surjective and its kernel is exactly the source-scaling line. Hence each normalized quotient solution is a reduced isolated point after quotienting by source scaling.*

Proof. The exact coefficient matrices have the following ranks and certified nonzero maximal minors.

map	(deg p , deg q)	rank $\mathcal{L}$	nonzero maximal minor
F_2 , 125	(1, 2)	2	$-36/5$
one-step 126	(3, 5)	7	2090188800
two-step 126	(4, 3)	6	37791360/7
F_{24} , $-$	(2, 3)	4	$99/20 - 153\sqrt{6}/40$
F_{24} , $+$	(2, 3)	4	$99/20 + 153\sqrt{6}/40$

In every row the domain dimension is one more than the displayed rank. The vector (up', uq') is nonzero and lies in the kernel by differentiating source scaling. It therefore spans the kernel. $\square$

Remark 3.4. Proposition 3.3 is a coefficient-level statement for the reduced cover. It does not prove that a full boundary deformation functor splits étale-locally into this Hurwitz point and an independent normal-jet problem.

4. THE NORMAL-BOUNDARY DETERMINANT COMPLEX

Let E be a smooth boundary curve with local coordinate z , and let s be a normal parameter. For positive integers α, β , consider

$$\alpha AB_z - \beta A_z B + s(A_z B_s - A_s B_z) = \Psi(z), \quad (4.1)$$

where

$$A = A_0 + \sum_{r \geq 1} s^r a_r, \quad B = B_0 + \sum_{r \geq 1} s^r b_r.$$

The order-zero equation is

$$\alpha A_0 B'_0 - \beta A'_0 B_0 = \Psi.$$

Proposition 4.1 (Universal layer operator). *At normal order r , the terms involving the new pair (a_r, b_r) are*

$$\boxed{\mathcal{D}_r^{\alpha, \beta}(a, b) = (\alpha - r)a dB_0 - \beta B_0 da + \alpha A_0 db + (r - \beta)b dA_0.} \quad (4.2)$$

Every other term at that order depends only on lower layers.

Proof. The coefficient of s^r in the first two terms of (4.1) contributes

$$\alpha(aB'_0 + A_0 b') - \beta(a'B_0 + A'_0 b).$$

The new part of the last term is $rA'_0 b - rA_0 b'$. Combining these terms and multiplying by dz gives (4.2). $\square$

Proposition 4.2 (Residue adjoint). *For a meromorphic scalar λ ,*

$$\boxed{(\mathcal{D}_r^{\alpha, \beta})^\vee(\lambda) = (\beta B_0 d\lambda + (\alpha + \beta - r)\lambda dB_0, -\alpha A_0 d\lambda + (r - \alpha - \beta)\lambda dA_0).} \quad (4.3)$$

More precisely,

$$\begin{aligned} \lambda \mathcal{D}_r(a, b) - a((\mathcal{D}_r)^\vee \lambda)_1 - b((\mathcal{D}_r)^\vee \lambda)_2 \\ = d(-\beta \lambda B_0 a + \alpha \lambda A_0 b). \end{aligned} \quad (4.4)$$

Consequently every adjoint solution in a dual pole window gives a residue functional annihilating the image of $\mathcal{D}_r$.

Proof. Expand the right side of (4.4). Summing residues on a complete curve kills the exact differential. $\square$

If the chosen rational-differential window represents the full dual of the target space under the residue pairing, the kernel of (4.3) represents the entire cokernel. This is the precise finite-window setting in which a left null vector of a monomial coefficient matrix becomes an intrinsic residue obstruction.

Proposition 4.3 (Virtual obstruction excess on $\mathbb{P}^1$). *Suppose*

$$\mathcal{D}_r : H^0(\mathbb{P}^1, \mathcal{O}(d_A)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(d_B)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(d_W)),$$

with $d_A, d_B, d_W \geq -1$. The target-minus-domain dimension is

$$\epsilon = d_W - d_A - d_B - 1. \quad (4.5)$$

If $\mathcal{D}_r$ is injective, its cokernel has dimension ϵ .

Proof. Use $h^0(\mathcal{O}(d)) = d + 1$ for $d \geq -1$. $\square$

For the independently certified full $(8, 28)$ support, the windows at $r \geq 5$ are

$$d_A = 10 - r, \quad d_B = 15 - r, \quad d_W = 25 - r.$$

Thus

$$\epsilon_r = r - 1.$$

This numerical specialization is evidence for the usefulness of the complex, but it is not used to claim the public degree-125 theorem.

5. A UNIVERSAL SECONDARY BELYI MAP

Let $D = \{t = 0\}$ be a boundary component with coordinate s , and put

$$c(s) = \frac{s}{s-1}.$$

Assume target coordinates have been normalized to

$$\pi = t^a c(s) + O(t^{a+1}), \quad u = \sum_{j \geq 1} t^j h_j(s), \quad (5.1)$$

where $a \geq 1$. Suppose the boundary-volume identity has no term below order t^{a+e-1} and that its first nonzero coefficient is

$$d\pi \wedge du = -t^{a+e-1}(s-1)^{-a-2} ds \wedge dt + O(t^{a+e}). \quad (5.2)$$

At a lower order j , the homogeneous equation is

$$jc'h_j - ach'_j = 0. \quad (5.3)$$

Lemma 5.1 (Resonance and target shears). *A nonzero rational solution of (5.3) exists if and only if $a \mid j$. If $j = ak$, every solution is $h_j = Cc^k$, and its leading contribution $Ct^{ak}c^k$ is removed by the target shear $u \mapsto u - C\pi^k$.*

Proof. Equation (5.3) gives $h_j = Cc^{j/a}$. Since $\text{div}(c) = (0) - (1)$, the power is rational exactly when j/a is an integer. The assertion about the shear follows from (5.1). $\square$

After removing all resonant lower terms, the coefficient of (5.2) gives

$$ec'h - ach' = -(s-1)^{-a-2}. \quad (5.4)$$

5.1. The contact degree.

Proposition 5.2 (Contact-degree formula). *Suppose coprime integers m, n satisfy*

$$v_E(P) = -m, \quad v_E(Q) = -n, \quad v_D(P) = -ma, \quad v_D(Q) = -na.$$

Choose $c, d \in \mathbb{Z}$ with $dn - cm = 1$ and set

$$\pi = P^c/Q^d, \quad \tau = Q^m/P^n.$$

Assume that in the toric chart $x = t^{-1}$ up to a unit and that

$$dP \wedge dQ = x^\kappa dx \wedge dy$$

up to a nonzero s -unit. Then the first possible non-shear contact degree is

$$e = a(m + n) - \kappa - 1. \quad (5.5)$$

Proof. The Bézout relation gives $v_D(\pi) = a$ and $v_D(\tau) = 0$. Direct differentiation gives

$$d\pi \wedge d\tau = -P^{c-n-1}Q^{m-d-1}dP \wedge dQ.$$

The monomial factor has t -order $a(m + n + 1)$, while the last wedge has order $-\kappa - 2$. On the other hand, the wedge of $\pi \sim t^a$ with $\tau - \tau|_D \sim t^e$ has order $a + e - 1$. Equating orders yields (5.5). $\square$

5.2. Transport and passport.

Theorem 5.3 (Universal secondary-boundary transport). *Let $a, e \in \mathbb{N}$ satisfy $e > a^2$, and put $\delta = e - a^2$. Equation (5.4) has a unique solution*

$$h(s) = \frac{H_{a,e}(s)}{(s-1)^a}$$

whose numerator is a polynomial of degree at most a , where

$$H_{a,e}(s) = \sum_{k=0}^a \frac{a^k a!}{(a-k)! \prod_{j=0}^k (e - aj)} s^k \quad (5.6)$$

$$= \frac{1}{e} {}_2F_1\left(-a, 1; 1 - \frac{e}{a}; s\right). \quad (5.7)$$

It satisfies

$$\begin{aligned} as(s-1)H' + (e - a^2s)H &= 1, \\ H(0) &= \frac{1}{e}, \quad H(1) = \frac{1}{\delta}, \end{aligned} \quad (5.8)$$

and H is squarefree.

The exceptional ratio

$$W_{a,e}(s) = \frac{(s-1)^\delta H_{a,e}(s)^a}{s^e} \quad (5.9)$$

is a degree- e Belyi map. It obeys

$$W'_{a,e}(s) = \frac{(s-1)^{\delta-1} H_{a,e}(s)^{a-1}}{s^{e+1}}, \quad (5.10)$$

and its passport is

$$(a^a, \delta), \quad (e), \quad (a+1, 1^{e-a-1}). \quad (5.11)$$

Proof. Substitute $h = H/(s-1)^a$ into (5.4); since $c' = -(s-1)^{-2}$, this gives (5.8). Writing $H = \sum h_k s^k$ gives

$$h_0 = \frac{1}{e}, \quad h_k = \frac{a(a+1-k)}{e - ak} h_{k-1}.$$

The denominators are nonzero because $e > a^2$, and the recurrence gives (5.6). It also proves uniqueness. The hypergeometric notation is the same terminating recurrence. Evaluation at zero and one gives the stated endpoint

values. If $H(s_0) = 0$, then (5.8) gives $as_0(s_0 - 1)H'(s_0) = 1$; hence every root is simple and avoids 0, 1.

Logarithmic differentiation of (5.9), followed by (5.8), gives

$$\frac{W'}{W} = \frac{1}{s(s-1)H},$$

which is (5.10). The numerator and denominator of W are coprime of degree e . Over zero, the a roots of H have multiplicity a and $s = 1$ has multiplicity δ . Over infinity, $s = 0$ has multiplicity e . At $s = \infty$, (5.10) shows that $W - W(\infty)$ has order $a + 1$. Finally,

$$a(a-1) + (\delta-1) + (e-1) + a = 2e-2,$$

so Riemann–Hurwitz leaves only $e - a - 1$ unramified points in the third fiber and proves (5.11). $\square$

Corollary 5.4 (The $(4, 17)$ cover). *For $a = 4, e = 17$,*

$$H_{4,17}(s) = \frac{195 + 240s + 320s^2 + 512s^3 + 2048s^4}{3315},$$

$$W_{4,17}(s) = \frac{(s-1)H_{4,17}(s)^4}{s^{17}}, \quad W'_{4,17}(s) = \frac{H_{4,17}(s)^3}{s^{18}},$$

with passport $(4^4, 1), (17), (5, 1^{12})$.

6. FILTERED RESIDUES AND THE REMAINING GLUING PROBLEM

The phrase “vanishing high-pole residues lowers the Newton window” has an exact linear core.

Lemma 6.1 (Filtered one-layer descent). *Let $D : U \rightarrow V$ be a linear map of finite-dimensional vector spaces and let $V_{\leq d} \subseteq V$ be a filtered subspace. For $\Phi \in V$, the following are equivalent:*

- (i) *there exists $u \in U$ such that $\Phi - Du \in V_{\leq d}$;*
- (ii) *every*

$$\lambda \in (\text{im } D)^\perp \cap (V_{\leq d})^\perp$$

satisfies $\lambda(\Phi) = 0$.

Proof. Let $q : V \rightarrow V/\text{im } D$. Condition (i) says $q(\Phi) \in q(V_{\leq d})$. Finite-dimensional duality says precisely that every functional on the quotient vanishing on $q(V_{\leq d})$ must also vanish on $q(\Phi)$. Pulling the functionals back to V gives (ii). $\square$

For $\mathcal{D}_r$, the relevant functionals are the pole-filtered adjoint residue classes of proposition 4.2. Thus at one fixed normal layer their vanishing is equivalent to reducing the forcing term to the smaller pole window.

This is not yet a descent theorem for polynomial Keller pairs. The corrections must be chosen compatibly at all layers, and the resulting formal transformation must integrate to an allowable polynomial approximate-root operation.

Question 6.2 (Integrable filtered descent). *For a terminal complete-chain boundary model, does simultaneous vanishing of all high-pole residue obstructions either force the boundary-gluing scheme to be empty or produce a polynomial transformation of strictly smaller complete-chain complexity?*

A positive answer, together with completeness of the finite boundary model, would permit induction on complete-chain complexity. Neither hypothesis is proved here.

Question 6.3 (Finite determinacy and algebraization). *For a fixed complete Newton chain, do sufficiently many mutually compatible normal layers force a finite polynomial model? Equivalently, can compatible formal solutions continue to all orders without algebraizing to an allowable polynomial transformation?*

Question 6.4 (Normal splitting). *Near one of the rigid reduced covers of proposition 3.3, does the complete boundary deformation functor split étale-locally into the reduced Hurwitz point and a normal-jet deformation problem? If not, which cross-terms measure the failure?*

7. EXACT COMPUTATION AND REPRODUCIBILITY

All finite computations in this paper were rerun with exact arithmetic. The certificate suite performs the following checks.

- (1) It verifies the layer operator, the adjoint integration-by-parts identity, and the specialization $\epsilon_r = r - 1$.
- (2) For $1 \leq a \leq 8$ and $\delta \in \{1, 2, 3\}$, it verifies the hypergeometric recurrence, squarefreeness, derivative identity, degrees, passports, and Riemann–Hurwitz count for $e = a^2 + \delta$.
- (3) It verifies the terminal-corner arithmetic, lattice divisibilities, quotient equations, derivative formulas, and the explicit degree-6 map.
- (4) It evaluates the exact symmetric-group character sums, recovers the eleven ambient degree-30 classes, and selects the unique lattice-compatible C_5 pullback.
- (5) It verifies all five quotient passports and Hurwitz counts and all explicit quotient maps through maximum degree 128.
- (6) It constructs the five exact tangent matrices, proves their ranks, checks their scaling kernels, and records a nonzero maximal minor in each case.

Freshly generated JSON outputs are byte-for-byte identical to the recovered reference outputs. These scripts verify the explicit identities and finite enumerations. They do not verify that the complete-chain input exhausts every hypothetical counterexample, prove the missing gluing theorem, or establish novelty.

APPENDIX A. GAUGE NORMAL FORM AND FILTERED RESIDUE ADJOINTS

The normal-layer operator of section 4 has a simpler intrinsic form. This section separates field-level solvability from the finite Newton-window obstructions used in computation.

Write

$$A = A_0, \quad B = B_0, \quad S = \alpha + \beta,$$

and define the boundary one-form

$$\Omega = \alpha A dB - \beta B dA.$$

For a layer variation (a, b) , put

$$u = \frac{a}{A}, \quad v = \frac{b}{B}, \quad \xi = u + v, \quad \eta = \alpha v - \beta u.$$

Proposition A.1 (Gauge normal form). *The universal layer operator satisfies*

$$\mathcal{D}_r(a, b) = AB \nabla_r \eta + \frac{S-r}{S} \xi \Omega, \quad \nabla_r = d + \frac{r}{S} d \log(AB).$$

If $r \neq S$ and $\Omega \neq 0$, the operator is surjective over the function field. At the resonant layer $r = S$,

$$\mathcal{D}_S(a, b) = d(\alpha Ab - \beta Ba).$$

Proof. Substitute

$$u = \frac{\alpha \xi - \eta}{S}, \quad v = \frac{\beta \xi + \eta}{S}$$

into the coordinate formula for $\mathcal{D}_r$ and collect the ξ - and η -terms. Off resonance, for a rational differential Φ , set $\eta = 0$ and

$$\xi = \frac{S}{S-r} \frac{\Phi}{\Omega}.$$

At $r = S$, the multiplier term vanishes and $AB \nabla_S \eta = d(AB \eta)$. $\square$

Thus, away from resonance, a cokernel can only be created by the finite pole and Newton windows imposed on a , b , and the target differential. It is not a local obstruction to solving the rational differential equation.

Proposition A.2 (Extended deformation complex). *Allow a variation $\rho \Omega$ of the normalized right-hand side and put*

$$\tilde{\mathcal{D}}_r(a, b, \rho) = \mathcal{D}_r(a, b) - \rho \Omega.$$

Then

$$G_r(h) = (\alpha Ah, \beta Bh, (S-r)h)$$

satisfies

$$\tilde{\mathcal{D}}_r \circ G_r = 0.$$

Consequently the normal layer carries a complex

$$\mathcal{G}_r \xrightarrow{G_r} \mathcal{U}_{A,r} \oplus \mathcal{U}_{B,r} \oplus \mathcal{R}_r \xrightarrow{\tilde{\mathcal{D}}_r} \mathcal{W}_r,$$

whose degree $-1, 0, 1$ terms represent infinitesimal gauge, deformations modulo gauge, and obstructions.

Proof. The normal form gives

$$\mathcal{D}_r(\alpha Ah, \beta Bh) = (S - r)h\Omega.$$

□

The same normal form clarifies the adjoint. If $M = AB$, integration by parts gives

$$\lambda \mathcal{D}_r(a, b) = d(\lambda M \eta) - M \eta \nabla_{S-r} \lambda + \frac{S-r}{S} \lambda \xi \Omega.$$

Thus

$$(M \nabla_r)^\dagger = -M \nabla_{S-r}.$$

The complementary-layer symmetry $r \leftrightarrow S - r$ is invisible in the original four-term coordinate formula.

Theorem A.3 (Filtered residue adjoint). *Let $U_{A,r}, U_{B,r} \subset k(E)$ be the permitted correction windows and let W_r be the permitted target window. Represent $W_r^\vee$ by principal parts $\lambda = (\lambda_p)$ at the allowed poles, with pairing*

$$\langle \lambda, \omega \rangle = \sum_p \text{Res}_p(\lambda_p \omega).$$

Then

$$(\text{coker } \mathcal{D}_r)^\vee \cong \ker \mathcal{D}_{r,\text{win}}^\vee,$$

where the filtered adjoint condition is that

$$\beta B d\lambda + (S - r)\lambda dB$$

annihilate $U_{A,r}$, and

$$-\alpha A d\lambda + (r - S)\lambda dA$$

annihilate $U_{B,r}$. The equation

$$\mathcal{D}_r(a_r, b_r) = -\Phi_r$$

is solvable exactly when

$$\sum_p \text{Res}_p(\lambda_p \Phi_r) = 0$$

for every filtered adjoint class.

Proof. Multiply the coordinate operator by λ , integrate each derivative term by parts, and apply the residue theorem. Boundary terms are precisely the displayed principal-part pairings. Finite-dimensional duality then identifies the cokernel dual with the filtered adjoint kernel. □

There is an exact coefficient dictionary. For target basis

$$dz, z dz, \dots, z^d dz$$

and a left-null vector $c = (c_0, \dots, c_d)$, set

$$\lambda_c(z) = \sum_{j=0}^d c_j z^{-j-1}.$$

Then

$$\text{Res}_{z=0} \left(\lambda_c \sum_{j=0}^d f_j z^j dz \right) = \sum_{j=0}^d c_j f_j.$$

Thus a matrix null vector and its principal-part representative are exactly the same functional in two coordinate systems.

Corollary A.4 (Resonant de Rham obstruction). *On $E = \mathbb{P}^1$, let Σ be the pole set and suppose the allowed primitive window is complete modulo constants. At $r = S$,*

$$\text{coker } \mathcal{D}_S \cong H_{\text{dR}}^1(\mathbb{P}^1 \setminus \Sigma) \cong \left\{ (\rho_p)_{p \in \Sigma} : \sum_p \rho_p = 0 \right\}.$$

In particular, its dimension is $|\Sigma| - 1$.

Proof. At resonance the image consists of exact differentials. A rational differential on $\mathbb{P}^1$ has a rational primitive exactly when all residues vanish, and the residue theorem supplies the single relation among them. $\square$

Status of the six terminal components. The preceding results prove the operator normal form, gauge complex, and filtered residue realization abstractly. They do not identify the six terminal obstruction polynomials with six residues. That final identification requires the archived specialization (A_0, B_0, Φ_r) , the exact pole windows, and the six left-null vectors. The precise test is whether the decisive layer is resonant, has seven poles, and has a complete primitive window. Until that artifact-level comparison is performed, the “six residues” interpretation remains a concrete conjectural compression of the certificate rather than a theorem.

APPENDIX B. DIRECT DESCENT, LOCAL PUISEUX FLEXIBILITY, AND INCIDENCE SURFACES

This appendix collects three preliminary lessons that motivate the global boundary-cover methods in the body. The broad failure of direct affine slices, graph pullbacks, and target projections was established publicly by Bartosz Naskręcki [8]. We therefore credit the general direct-descent result to Naskręcki. The first calculation below is retained only as an explicit explanation of the dimensional mechanism.

B.1. Where the third variable enters. In the chosen-root coordinates (t, r, c) for the three-dimensional map, direct differentiation gives

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2} = \frac{1}{x}, \quad \det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x.$$

The second determinant factors further as

$$\det \frac{\partial(t, r)}{\partial(x, y)} = \frac{2}{x^2}, \quad \frac{\partial c}{\partial z} = -x^3.$$

Thus the constant Jacobian is the exact cancellation

$$\frac{1}{x} \cdot \frac{2}{x^2} \cdot (-x^3) = -2.$$

The marked-root incidence contributes a pole, the two plane root coordinates contribute two more powers of that pole, and the transverse coordinate supplies precisely the cancelling zero.

Fixing $c = \lambda$ makes the obstruction visible. If $\lambda \neq 0$, the source surface is

$$x(2 - 3xy - x^2z) = \lambda$$

and hence isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$; its map to the (a, b) -plane has Jacobian

$$[a_\lambda, b_\lambda]_{x,y} = \frac{2}{x^3},$$

a nonconstant unit. If $\lambda = 0$, the source splits as

$$\mathbb{A}^2 \sqcup (\mathbb{G}_m \times \mathbb{A}^1),$$

and the affine-plane sheet is the triangular automorphism

$$(y, z) \mapsto (z + 4y^2, y).$$

This is a concrete instance of Naskręcki's broader no-go: the obvious slice is either punctured or trivial.

B.2. The exact Puiseux recurrence. Let a finite-dicritical chart have the form

$$x = t^{-m}, \quad y = \sum_{j=1}^{K-1} a_j t^{-m+j} + \xi t^{-m+K}, \quad \Delta = K - 2m > 0.$$

Then

$$\det \frac{\partial(x, y)}{\partial(t, \xi)} = -m t^{\Delta-1}.$$

If a plane map has Jacobian determinant κ and

$$F \circ \Phi(t, \xi) = \sum_{\nu \geq 0} f_\nu(\xi) t^\nu,$$

coefficient comparison gives

$$\sum_{\substack{\nu + \mu = n \\ \nu \geq 1}} \nu \det(f_\nu, f'_\mu) = -m\kappa \mathbf{1}_{n=\Delta}. \quad (\text{B.1})$$

Theorem B.1 (Exact two-level exclusion). *Under the finite-dicritical hypotheses used above, an exactly terminating two-level expansion*

$$F \circ \Phi = f_0 + t^\Delta f_1$$

forces the limiting nonproperness component to be a smooth embedded $\mathbb{A}^1$. It therefore cannot occur for a plane Keller counterexample.

Proof. The two nonzero orders in (B.1) give

$$\det(f_1, f'_0) = -\frac{m\kappa}{\Delta}, \quad \det(f_1, f'_1) = 0.$$

The second identity makes the nonzero vector polynomial f_1 point in a constant direction. The first then supplies a polynomial Bézout identity for the derivative of f_0 , so the limiting parametrization is immersive and injective after the standard finite-dicritical normalization. Its normalization is $\mathbb{A}^1$, and the constant determinant identity makes it a smooth embedded affine-line component. The standard plane nonproperness-curve restrictions exclude such a component for a counterexample. $\square$

Remark B.2. The theorem excludes exact two-level termination, not an infinite expansion with two leading characteristic scales. It should not be confused with the older phrase “two characteristic pairs.”

B.3. No finite local obstruction. Write $s = t^\Delta$ and

$$G(s, \xi) = \sum_{n \geq 0} s^n f_n(\xi).$$

The recurrence becomes

$$\sum_{\substack{i+j=N \\ i \geq 1}} i \det(f_i, f'_j) = c \delta_{N1}.$$

For $f_0 = (p_0, q_0)$, the map

$$L_{f_0}(h) = \det(h, f'_0)$$

is surjective on $\mathbb{C}[\xi]^2 \rightarrow \mathbb{C}[\xi]$ precisely when $\gcd(p'_0, q'_0) = 1$. At order N , the only occurrence of f_N is $N L_{f_0}(f_N)$. Hence every finite jet over an immersive limit curve extends by one order.

Proposition B.3 (Nodal all-orders solution). *For*

$$f_0(\xi) = (\xi^2, \xi^3 - \xi), \quad u(\xi) = \left(-1, -\frac{3}{2}\xi\right),$$

one has

$$\det(u, f'_0) = 1, \quad \det(u, u') = \frac{3}{2}.$$

For every c , the algebraic series

$$A(s) = \frac{2}{3} \left(\sqrt{1 + 3cs} - 1 \right)$$

gives

$$G(s, \xi) = f_0(\xi) + A(s)u(\xi), \quad \det(G_s, G_\xi) = c.$$

Proof. The two determinant identities are immediate. Since

$$A'(s) \left(1 + \frac{3}{2} A(s) \right) = c,$$

bilinearity of the determinant gives the final identity. $\square$

Thus no obstruction involving only finitely many local Puiseux coefficients can settle the nodal case. A successful obstruction must use polynomial termination, finite Newton support, compatibility among boundary charts, or global geometry.

B.4. A corrected log-geometric exclusion. Let

$$X \subset \mathbb{P}^2 \times \mathbb{P}^1$$

be a smooth finite root-incidence hypersurface of bidegree (b, d) . Let H and L be the pullbacks of the hyperplane classes. Assume that the root projection is finite, its ramification divisor is reduced, no ramification component lies over the parameter line at infinity, and the reduced boundary is simple normal crossing. Then

$$K_X = (b - 3)H + (d - 2)L, \quad R = bH + (d - 2)L.$$

The simple-root affine locus removes the pullback of parameter infinity and the ramification divisor, so for its reduced boundary D ,

$$K_X + D = (2b - 2)H + 2(d - 2)L.$$

Theorem B.4 (Squarefree-at-infinity incidence exclusion). *Under the preceding hypotheses, if $b, d \geq 2$, the affine simple-root locus is not isomorphic to $\mathbb{A}^2$.*

Proof. The class H is ample and L is nef, so $K_X + D$ is big (indeed ample in the finite setting). The open surface is of log general type, whereas $\mathbb{A}^2$ has logarithmic Kodaira dimension $-\infty$. $\square$

Remark B.5 (Correction). The hypothesis at parameter infinity is essential. When ramification lies there, the same divisor occurs in both the pullback of infinity and the ramification divisor but only once in the reduced boundary. The unrestricted earlier log-general-type claim double-counted this component and is withdrawn; only theorem B.4, or a version with an explicit log-canonical boundary hypothesis, is retained.

APPENDIX C. THE DEGREE-21 DIVISOR AND TERMINAL CERTIFICATES

The body develops the lattice-gap quotient covers. Here we record the complementary calculation on the original lower face and the exact certificates obtained after substituting its five dessins into the two surviving Newton supports.

C.1. The lower face. For either surviving support in the reduction of Gucione et al. [7], the transformed bracket is

$$[P, Q] = X^2.$$

For the primitive toric valuation

$$\nu(X) = -2, \quad \nu(Y) = 1,$$

put $z = XY^2$. The lower faces have the form

$$P_{\text{face}} = Xp(z), \quad Q_{\text{face}} = X^2Yq(z),$$

where $\deg p = 7$, $\deg q = 10$, and

$$pq + 2zpq' - 3zp'q = 1. \tag{C.1}$$

Proposition C.1 (The degree-21 Belyi map). *The rational function*

$$\tau(z) = z \frac{q(z)^2}{p(z)^3}$$

is a degree-21 Belyi map with passport

$$(2^{10}1), \quad (3^7), \quad (171^4).$$

There are exactly five connected dessin isomorphism classes with this passport. They form one arithmetic orbit over an irreducible quintic field and have monodromy A_{21} .

Proof. Equation (C.1) implies that p, q are coprime and nonzero at zero. Differentiation gives

$$\tau'(z) = \frac{q(z)}{p(z)^4}.$$

Thus the ten roots of q give double zeros, the seven roots of p give triple poles, and infinity has ramification index 17. The total ramification is

$$10 + 14 + 16 = 40 = 2 \cdot 21 - 2,$$

so there is no additional branch value.

The exact Murnaghan–Nakayama character calculation in S_{21} , followed by the class-size factor, gives $5 \cdot 21!$ labeled triples. Each is transitive, and the deck group is trivial, so there are five isomorphism classes. Exact coefficient reconstruction places all five over one quintic field; exact permutation certificates give monodromy A_{21} . The enumeration and reconstruction are computer-assisted. $\square$

C.2. Identification with Borisov’s divisor.

Theorem C.2 (Valuation over F_{-5}). *Let E_ν be the source divisor defined by ν . Its target center is Borisov’s divisor F_{-5} . The normal ramification index and residue degree are*

$$(e, f) = (1, 21).$$

Consequently neither surviving Newton support can realize Borisov’s Three-dessin framework.

Proof. Set $t = Y$, so $x = t^2/z$ in the original source coordinates. The source volume form is

$$dx \wedge dy = t^{-6} z^2 dt \wedge dz.$$

After adding the reduced boundary coefficient, the augmented-canonical label is -5 .

The face expansions are

$$P = t^{-2} z p(z) + \text{higher } \nu\text{-terms}, \quad Q = t^{-3} z^2 q(z) + \text{higher } \nu\text{-terms}.$$

On the target put

$$s = \frac{P}{Q}, \quad \lambda = \frac{Q^2}{P^3}.$$

Then s is a uniformizer and

$$dP \wedge dQ = -s^{-6} \lambda^{-3} ds \wedge d\lambda,$$

so the target label is also -5 . Pullback gives

$$\frac{P}{Q} = t \left(\frac{p(z)}{z q(z)} + O(t) \right),$$

hence $e = 1$. The residue of λ is

$$\text{res}_{E_\nu}(\lambda) = z \frac{q(z)^2}{p(z)^3} = \tau(z),$$

so $f = 21$ by proposition C.1.

In Borisov's target coordinates $(y_1, y_2) = (Q, P)$, the divisor F_{-5} is characterized by valuations $(-3, -2)$ and by the nonconstant residual parameter y_1^2/y_2^3 . Those are exactly the values above. Later infinitely-near divisors with the same two numerical orders have constant residue of this parameter, so they are distinguished.

Borisov's Three-dessin framework gives residue degree 16 on its unique source divisor above F_{-5} [4]. The forced divisor has residue degree 21, a contradiction. This excludes the named framework, not the two supports themselves. $\square$

Corollary C.3. *Any plane counterexample of maximum coordinate degree below 125 has generic degree at least 21.*

Proof. The published below-125 reduction leaves the two supports above, and the generic-degree formula contains the contribution $ef = 21$ from theorem C.2. $\square$

Question C.4. *Is E_ν the only source divisor over the generic point of F_{-5} ? The calculation $f = 21$ is one residue-degree contribution; cancellation valuations or infinitely-near divisors could contribute additional sheets.*

C.3. Exact support certificates. Substitute each of the five reconstructed lower faces into the truncated Newton support. This gives seven compatibility equations over the quintic field. A nonzero exact resultant, certified by good reduction, eliminates every dessin. Thus the truncated support is empty in characteristic zero.

For the full support, exact layer elimination and normalization reduce the problem to fifteen equations in five variables over the same quintic field. Six selected equations already generate the unit ideal.

Theorem C.5 (Terminal unit-ideal certificate). *For the stored normalized full-support system, the six selected obstruction equations generate the unit ideal over the quintic coefficient field.*

Exact computer-assisted certificate. The Macaulay multiplication map has 10,824 columns. The certificate specifies 7,121 rows and 7,121 pivot columns. Reducing this exact minor at a good prime and at the selected quintic embedding gives

$$\det = 859 \pmod{2053}.$$

The determinant is therefore nonzero in characteristic zero. The certificate package regenerates all fifteen exact equations, rebuilds the matrix, and replays the fixed pivot minor. $\square$

Combining the truncated resultant and theorem C.5 eliminates both encoded support alternatives after the degree-21 face substitution. This statement is exact for the stored systems. Turning it into a stand-alone global below-125 proof requires a line-by-line audit that the published Newton reduction, the five faces, all saturations and normalizations, and the layer systems exhaust every candidate.

There is also a smaller geometric compression. Five partial-gluing equations have mixed volume 296; on the certified 296-dimensional modular finite algebra, the sixth obstruction is a unit, and its rational norm over the five split embeddings is

$$51 \pmod{2053}.$$

This is already a compact representative of the large certificate. To use it as an independent characteristic-zero proof, one still needs a publication-level toric-saturation lemma showing that the finite étale special fiber exhausts the complete toric intersection.

C.4. Credit for the 125 bound. MathOverflow user `ratto3423` publicly announced a lower bound of 125 for the maximum coordinate degree of any characteristic-zero plane counterexample. The claim rests on a computer-assisted elimination of the same two supports [9]. We credit the public theorem announcement to `ratto3423`. The answer does not disclose enough method to compare its computation with the certificates above.

Accordingly, our claim is not priority for the degree-125 conclusion. The potential contribution of this project is the explicit degree-21 boundary description and an independently auditable terminal certificate architecture.

The current certificate is conditional only at the upstream exhaustiveness layer described above, not at its final linear-algebra step.

APPENDIX D. THE TERMINAL BOUNDARY OF THE PARAMETRIC F_2 FAMILY

The complete-chain F_2 family has

$$A_0 = (5, 20), \quad A'_0 = (1, 0), \quad A_1 = \left(\frac{7}{5}, 2\right),$$

$$n = 2m - 1, \quad m \geq 2,$$

and coordinate-degree pair

$$(25m, 25(2m - 1)).$$

This appendix classifies the final face for every m , shows that its local defect equations are integrable, and gives a global no-go theorem for the natural five-band lift. It does not eliminate the full F_2 family.

D.1. The final regular corner. If (ρ, σ) is the direction at A_1 , the type-I corner identity gives

$$(7m - 4)\rho + (10m - 5)\sigma = 0.$$

With

$$d_m = \gcd(10m - 5, 7m - 4) \in \{1, 5\},$$

the primitive direction is

$$(\rho_m, \sigma_m) = \left(\frac{10m - 5}{d_m}, -\frac{7m - 4}{d_m}\right).$$

The type-I.a start formula would require the positive integer

$$a' = \frac{1}{2m - 1},$$

and is therefore impossible. Type I.b has $k = 1$ and starts

$$\text{st } P_E = \left(\frac{4}{5}, 1\right), \quad \text{st } Q_E = \left(\frac{1}{5}, 0\right).$$

Set

$$Z = x^{(7m-4)/5} y^{2m-1}.$$

After source and target scaling, the final leading forms are

$$P_E = x^{4/5} y p(Z), \quad p(Z) = Z - 1,$$

$$Q_E = x^{1/5} q(Z), \quad \deg q = 2.$$

Their bracket equation is

$$-pq + mZpq' - (2m - 1)Zp'q = \lambda. \tag{D.1}$$

Theorem D.1 (Final-face classification). *Equation (D.1) has the unique normalized solution*

$$q_m(Z) = Z^2 - \frac{2m-1}{m}Z + \frac{(2m-1)(m-1)}{2m^2},$$

$$\lambda_m = \frac{(2m-1)(m-1)}{2m^2}.$$

The discriminant is

$$\text{Disc}(q_m) = \frac{2m-1}{m^2} \neq 0.$$

Proof. Substitute $p = Z - 1$ and a monic quadratic $q = Z^2 + \alpha Z + \beta$ into (D.1). Vanishing of the coefficients of Z^2 and Z gives

$$\alpha = -\frac{2m-1}{m}, \quad \beta = \frac{(2m-1)(m-1)}{2m^2}.$$

The constant term is $\lambda_m = \beta$, and direct substitution gives the discriminant. $\square$

D.2. The alternating boundary dessin. Define

$$\beta_m(Z) = \frac{q_m(Z)^m}{Z(Z-1)^{2m-1}}.$$

Logarithmic differentiation using (D.1) gives

$$\beta'_m(Z) = \lambda_m \frac{q_m(Z)^{m-1}}{Z^2(Z-1)^{2m}}.$$

Theorem D.2 (Unique alternating dessin). *The map β_m is a degree- $2m$ Belyi map with passport*

$$(m, m), \quad (2m-1, 1), \quad (3, 1^{2m-3}).$$

It is the unique normalized map of this form, and its monodromy group is

$$\text{Mon}(\beta_m) = A_{2m}.$$

Proof. The derivative shows that the roots of q_m lie over 0, each with ramification m ; $Z = 1$ is a pole of order $2m-1$; $Z = 0$ is a simple pole; and $Z = \infty$ has ramification index 3. The total ramification is $4m - 2 = 2 \deg \beta_m - 2$, so there is no further ramification.

On $2m$ letters, take

$$\sigma_\infty = (1 \ 2 \ \cdots \ 2m-1), \quad \sigma_1 = (1 \ m \ 2m),$$

and $\sigma_0 = (\sigma_\infty \sigma_1)^{-1}$. Then σ_0 is the product of two m -cycles. The group is primitive because a block containing the fixed point of σ_∞ is either a singleton or the whole set. For $m \geq 3$, Jordan's theorem and the contained three-cycle give A_{2m} ; all generators are even. The case $m = 2$ is checked directly and gives A_4 .

Finally, the unique pole of order $2m-1$, the simple pole, and the index-three point normalize the source points to $1, 0, \infty$; theorem D.1 then gives uniqueness. $\square$

Corollary D.3 (Complete final-face package). *For every $m \geq 2$, the F_2 terminal corner has the unique normalized face polynomial q_m of theorem D.1. Its unique normalized degree- $2m$ Belyi map has passport*

$$(m, m), \quad (2m - 1, 1), \quad (3, 1^{2m-3})$$

and monodromy group A_{2m} .

Proposition D.4 (The $m = 3$ self-similarity). *For $m = 3$,*

$$q_3(Z) = Z^2 - \frac{5}{3}Z + \frac{5}{9},$$

and the change $Z = \frac{5}{3}t$ gives

$$\beta_3\left(\frac{5}{3}t\right) = \frac{(t^2 - t + 1/5)^3}{t(t - 3/5)^5}.$$

The right side is exactly the degree-six A_6 Type-II first-face dessin. This is a genuine self-similarity of the two boundary maps, but it does not identify their ambient boundary components and is not a gluing theorem.

D.3. Unimodular defect coordinates. Pass to $X = x^{1/5}$, translate the selected double root to $u = 0$, and put

$$Z = X^{5m-3}u^{2m-1}, \quad \epsilon = X^{-5}u^{-2}.$$

The exponent matrix has determinant

$$\det \begin{pmatrix} -5 & -2 \\ 5m-3 & 2m-1 \end{pmatrix} = -1,$$

so

$$X = \epsilon^{-(2m-1)}Z^{-2}, \quad u = \epsilon^{5m-3}Z^5.$$

Write

$$P = \epsilon^{-m}Z^{-1}A(\epsilon, Z), \quad Q = \epsilon^{-(2m-1)}Z^{-2}B(\epsilon, Z).$$

Then

$$[P, Q]_{x,y} = \frac{1}{5}\mathcal{K}_m(A, B),$$

where

$$\begin{aligned} \mathcal{K}_m(A, B) = & -AB + mZAB_Z - (2m-1)ZA_ZB \\ & + \epsilon[(ZA_Z - A)B_\epsilon + (2B - ZB_Z)A_\epsilon]. \end{aligned} \quad (\text{D.2})$$

Theorem D.5 (Local defect integrability). *For every $m \geq 2$, the local equation*

$$\mathcal{K}_m(A, B) = \lambda_m$$

has a nontrivial exact finite-support one-parameter family satisfying the Newton support conditions. Consequently no obstruction based only on finite-order local lifting at the final face can eliminate the F_2 family.

Proof. Use

$$A = Z - 1 + Za(\epsilon), \quad B = q_m(Z) + Zb(\epsilon) + Z^2c(\epsilon).$$

Equation (D.2) reduces to two ordinary differential equations. With free parameter τ , an exact solution is

$$\begin{aligned} a(\epsilon) &= \frac{2m(m-2)}{(2m-3)(2m-1)}\tau\epsilon, \\ b(\epsilon) &= -\frac{2(m-1)^2}{m(2m-3)}\tau\epsilon, \\ c(\epsilon) &= \tau\epsilon + \frac{4(m-2)(m-1)^3}{(2m-3)^3(2m-1)}\tau^2\epsilon^2. \end{aligned}$$

Direct substitution yields $\mathcal{K}_m(A, B) = \lambda_m$. $\square$

D.4. A global five-band no-go theorem. Before translating the root, put $v = Xy$. Polynomial descent is the invariant-ring condition for

$$(X, v) \mapsto (\zeta X, \zeta v), \quad \zeta^5 = 1.$$

The natural lift using only the bands forced directly by the first and final faces is

$$\begin{aligned} P &= X^{5m}R(v)^m + X^3U(v), \\ Q &= X^{5(2m-1)}R(v)^{2m-1} + X^{5m-2}V(v) + XW(v). \end{aligned}$$

Cyclic descent forces

$$U = v^2A(v^5), \quad V = v^2B(v^5), \quad W = v^4C(v^5).$$

Theorem D.6 (Five-band descent no-go). *No pair of the displayed five-band form descends to a plane polynomial map with nonzero constant Jacobian.*

Proof. Since

$$\det \frac{\partial(x, y)}{\partial(X, v)} = 5X^3,$$

a nonzero constant original Jacobian requires a nonzero constant coefficient at X^3 in the bracket

$$[P, Q]_{X, v}.$$

The only contribution from the five bands is

$$3UW' - U'W.$$

Writing $s = v^5$ and using the character forms gives

$$3UW' - U'W = 5v^5(2AC + 3sAC' - sA'C),$$

which is divisible by v^5 . It cannot be the required nonzero constant. $\square$

D.5. The global attachment problem. For a general descended pair

$$P = \sum_{\alpha} X^{\alpha} P_{\alpha}(v), \quad Q = \sum_{\beta} X^{\beta} Q_{\beta}(v),$$

the exact band equation is

$$\sum_{\alpha+\beta=r+1} (\alpha P_{\alpha} Q'_{\beta} - \beta P'_{\alpha} Q_{\beta}) = 5\kappa \delta_{r,3}.$$

A constant v^0 -term at X^3 can arise only from the band pairs

$$(\alpha, \beta) = (5, -1) \quad \text{or} \quad (-1, 5),$$

corresponding to crossed affine-linear x - and y -terms.

Question D.7 (Two-sided F_2 attachment). *Does any finite two-sided band system simultaneously realize the rigid first face, the final A_{2m} face, the cyclic character conditions, and the affine bands $5, -1$? Equivalently, can the locally integrable terminal model be attached to a global plane polynomial Keller pair?*

The local theorem and the five-band theorem show why this question cannot be replaced by either a higher-order one-sided Puiseux calculation or the most economical global ansatz.

APPENDIX E. A SIX-SHEET MONODROMY PROBLEM

The boundary calculations above suggest a separate problem at generic degree six.

Question E.1 (Six-sheet monodromy). *Must every hypothetical six-sheet counterexample to the plane Jacobian conjecture have monodromy group A_6 or S_6 ?*

Remark E.2 (Present evidence). A dated synthesis reported the corresponding necessary-condition statement and eliminations of several conditional compactification strata. The available archive does not contain the underlying proof or computational artifacts. We therefore record the monodromy statement as an open problem and do not claim the reported stratum eliminations as theorems.

AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. Human specialist review has not yet been completed. The evidence for the computer-assisted propositions is the exact certificate suite described above, not language-model output.

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